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Additive Manufacturing/3D Printing Potential in Oil and Gas Industry: A Roadmap to Efficiency and Customization

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Abstract

The oil and gas industry stands as one of the world's most capital-intensive and technologically complex sectors. Characterized by massive investments, intricate infrastructure, and extensive supply chains, this industry is the backbone of global energy production and distribution. From upstream exploration and extraction to downstream refining and distribution, every phase is governed by high stakes—both in terms of financial outlay and operational continuity.

One critical aspect often overlooked outside industry circles is the indispensable need for swift maintenance, particularly the rapid availability of replacement parts and components. Given the sheer scale of production and the enormous costs associated with downtime, the ability to perform quick and effective maintenance is not simply a matter of efficiency—it also mandates to explore the state of the art innovative techniques for manufacturing the complex maintenance parts involved in the machines used in such industry, which is a linchpin for avoiding catastrophic production losses and safeguarding the bottom line. One promising avenue is the use of 3D printing/Additive Manufacturing for the metallic spare parts used in oil and gas industry. This research paper explores the potential, challenges, and future roadmap for integrating 3D printing technology into the oil and gas sector, focusing on metallic parts' efficiency and customization

Introduction

Additive manufacturing (AM), commonly known as 3D printing, has emerged as a transformative force across a spectrum of industries, from aerospace to healthcare. In recent years, the oil and gas sector—traditionally characterized by heavy infrastructure, complex supply chains, and high operational risks—has initiated to explore the implementation of this technology. The promise: streamlined maintenance, accelerated innovation, and enhanced resilience in the face of global supply chain disruptions. However, integrating additive manufacturing into oil and gas operations is not without its challenges and limitations. This paper is based on the trial which examines the state of AM adoption, its potential benefits, and the obstacles that must be navigated to fully realize its promised benefits.

Additive manufacturing is a process that builds objects layer by layer from digital models. Unlike subtractive manufacturing—which removes material from a larger block—AM fabricates components by adding only the necessary material, often resulting in less waste and greater design flexibility. Technologies

within AM include fused deposition modeling (FDM), selective laser sintering (SLS), stereolithography (SLA), and direct metal laser sintering (DMLS), each with distinct materials and use cases.

Application of AM in the oil and gas industry is in the process of evolution, and it's establishing its gradual confidence amongst the global oil and Gas key operators Like ADNOC, Saudi Aramco, PDO, Exxon Mobil, Sinopec, CNPC, Shell etc. with the advent of standards, procedures and regulatory compliance requirements in this technology. Besides oil and Gas operators its being used at a functional level by many other industrial leaders like NASA, Boeing, Siemens etc. in the sectors of aerospace, automotive, R&D and numerous others. Main application areas in oil and Gas industry where AM integration can be implemented for unlocking the potential benefits associated to AM are as under:

- **Spare Parts-On demand Production:** One of the most promising uses of AM in oil and gas is the on-demand printing of replacement parts. This is especially valuable for legacy equipment, bespoke components, or obsolete parts where traditional supply chains may be slow or unreliable.
- **Tooling and Prototyping:** AM allows rapid prototyping and the creation of custom tools or jigs, accelerating innovation and reducing the lead time for new product introduction.
- **Complex Geometries:** Design freedoms offered by AM enable the creation of components with intricate internal structures, such as optimized flow channels or lattice reinforcements, which can enhance the performance and efficiency of pumps, valves, and other fluid-handling equipment.
- **Asset Life Extension:** The capability to produce discontinued or hard-to-source components gives operators the ability to extend the lifespan of existing machinery, reducing the need for costly system overhauls.
- **Design Optimization:** AM enables complex, customized structures for lighter, stronger, and more efficient parts than traditional methods.
- **Material Flexibility:** AM can utilize a wide variety of materials, including high-performance metals and polymers, allowing for customized part properties to suit demanding operating conditions.
- **Sustainable:** AM can help oil & gas, offshore, and maritime sectors reduce emissions by enabling distributed, just-in-time production, cutting transport needs and greenhouse gases. 3D printing creates less waste than traditional methods by adding only necessary material. Locating these processes near recycling facilities can support local circular economies.
- **Remote and Harsh Environments:** Offshore rigs and remote processing stations benefit from AM by reducing dependence on long and vulnerable supply chains, enabling the local production of critical spares.

This paper benchmarks the trial implementation of additive manufacturing for the manufacturing of metallic spare part, to show case that AM can play its role in enhancing competitiveness, operational efficiency, and sustainability. It describes the technical assessment based on adopting 3D printing for a critical spare part- oil recovery pump impeller with Duplex SS A890 grade 5A material, demonstrating its potential to address industry needs promptly and cost-effectively.

Regional initiatives for Additive Manufacturing Adoption

The United Arab Emirates (UAE) has been at the forefront, launching the "Dubai 3D Printing Strategy" in 2016, which aims to use 3D additive manufacturing for resolving critical challenges of oil and gas industry.

Additionally, Saudi Arabia's "Vision 2030" plan includes the development of a 3D printing industry to foster innovation and enhance local manufacturing capabilities.

Petroleum Development Oman (PDO), Oman's largest oil and gas company, has explored 3D printing for various uses. Partnering with Spare Parts 3D (SP3D), PDO assessed the 3D printability of 150,000 spare parts, identifying suitable candidates and developing a matrix to evaluate 3D printing feasibility for other components.

Saudi Arabia's National Industrial Strategy (NIS) recognizes the importance of AM and includes it in key manufacturing clusters, further driving its adoption across various sectors. Aramco collaborates with organizations like Baker Hughes/GE and supports the establishment of cutting-edge AM centers, such as the first industry 3D printer for metal in the Kingdom.

Egypt has promoted 3D printing through initiatives like the 3D Printing Center in Cairo. Similarly, Turkey supported the field by launching the Additive Manufacturing Technology Development Center in 2017.

Basis of Study

The intent of the study is rooted in the adoption of advanced 3D printing and additive manufacturing technologies, which have proven their effectiveness and are widely embraced within the oil and gas industry. Numerous operator companies have already implemented AM methods, producing essential components that have been reliably serving in the field for years. Additionally, there are several examples where field operators have utilized AM technology, through specialized vendors, to manufacture crucial parts such as pump and compressor impellers, valve plugs, retainers, control valve plug-cages, motor components, and various other maintenance items for both static and rotating machinery.

To utilize the technical and economical leverages of AM, the pilot study is performed for manufacturing of pump impeller, for necessary assessments prior to its implementation as proof of concept.

Further details about the trial 3D-printed part are listed as below:

Trial Part: Oil recovery pump (14-01-P-1101/1102), Impeller

Service: Crude Oil

Impeller Material: Duplex SS A890 grade 5A

Service Provider/Contractor: AM Vendor

The proposed change is justified as it is an integral part of the 3D-Printing initiative, aimed at leveraging advanced manufacturing technologies. When compared to traditionally manufactured impellers, the 3D printed part exhibited several advantages:

- **Design Flexibility:** The additive manufacturing process allowed for the creation of complex geometries that would be challenging to achieve with traditional methods.
- **Lead Time:** The production lead time for the 3D printed impeller is remarkably reduced, with a maximum turnaround of 1 month., enhancing the overall project timeline.
- **Economical:** The cost efficiency of 3D-Printing is significant, with parts being nearly 50% cheaper compared to those produced by conventional machining or casting methods.

The industry is currently navigating significant challenges, including oil price volatility and the transition toward alternative energy sources, making operational cost reduction an increasingly critical priority for sector stakeholders.

According to the World Economic Forum, additive manufacturing has the potential to deliver cost and time savings valued at up to \$30 billion for oil and gas companies.

- **In-Country Value (ICV):** Adoption of AM has potential to attract the local techno-entrepreneurs that will enhance the In-Country Value (ICV). All the manufacturing processes conducted for the trial part were within the UAE, thereby supporting local industry and incubating the culture of technological development within local techno environment.
- **Sustainability:** 3D-Printing is sustainable technology, and it results in fewer emissions, contributing to environmental sustainability.
- **Digital Inventory:** It also marks the initial step towards establishing a digital inventory for the Operator Company, which will modernize inventory management and release frozen capital in physical inventories.

One example of 3D printed burner designed for an industrial gas turbine is depicted as under. By utilizing additive manufacturing, the burner was produced as a single piece (shown on left) rather than being made in 13 separate parts that require welding (shown on Right).



Figure 1—3D-printed burners (Image credit: Siemens)

These benefits, along with several other advantages, underscore the importance and value of the trial.

Methodology

Integration Process: Opportunities and Strategies

The trial was based on the integration of AM to produce Oil recovery pump (14-01-P-1101/1102), Impeller, which demands a very strategic approach, and this process was accomplished by proceeding through the following main steps:

Material/3D Printing Method Selection

When selecting an appropriate metallic powder for 3D printing, it is essential to assess the mechanical requirements, cost considerations, and design complexities of the intended application relative to the powder's characteristics. The selection process is further influenced by the specific 3D printing technology employed, such as Powder Bed Fusion (PBF) or Directed Energy Deposition (DED).

For optimal and dependable printing results, it is advisable to evaluate the following essential powder characteristics:

1. Chemical composition: The precise elemental makeup of the powder determines the final part's mechanical properties, such as strength, ductility, and resistance to corrosion and heat.
2. Particle size distribution (PSD): The range of particle diameters affects powder flowability, packing density, surface finish, and final part density.
 - PBF (SLM/DMLS): Uses finer powders (typically 15–45 μm) for high resolution and detail.
 - EBM: Uses larger, coarser powders (typically 53–105 μm) due to its deeper melt pool.
 - DED: Is more tolerant of a wider range of particle sizes.

3. Particle morphology (shape): Highly spherical, smooth particles are preferred for PBF processes as they improve flowability, ensuring consistent powder layering. Non-spherical shapes can lead to poor packing density and defects.
4. Flowability and spread ability: These properties describe how well the powder spreads evenly across the build plate. They are critical for PBF and are influenced by particle shape, size, and moisture content.
5. Purity and cleanliness: Impurities, including oxygen content and contaminants, can significantly reduce the final part's mechanical properties. Using inert gas atomized powders helps maintain high purity.
6. Cost and availability: More specialized, higher-purity, and finer powders are generally more expensive. Weigh the material's cost against your application's performance requirements and budget.
7. Compatibility with the AM process: Not all powders are suitable for every additive manufacturing process. Ensure the powder's specifications (e.g., particle size) are compatible with your specific printer and technology.

Not all metals, polymers, or composites used in oil and gas are compatible with additive manufacturing. Many high-performance alloys for demanding environments remain unprintable with current AM technologies. A comparable challenge was encountered during the selection of impeller material: Inconel 625 was evaluated as a viable alternative to the original Duplex Stainless Steel A890 Grade 5A. A comparative analysis of both materials is presented below.

	Inconel 625	S.Duplex SS (Zetium 55)
Yield Strength Min. (MPa)	579.2	250
T.S. Min. (MPa)	992.9	500
Elongation Min. %	44	35
Hardness, HB	175 - 250	150 - 200
Cr%	20 - 23	20.5 - 21.5
Ni%	58 mn.	17.5 - 18.5
Mo%	8.0 - 10.0	5.0 - 5.4
N%		0.19 - 0.22
C%	0.1 max.	0.03 max.
Si%	0.5 max.	0.8 max.
Mn%	0.5 max.	1.0 max.
Calculated PREN, min.	46.4	40.04

Figure 2—Duplex SS Vs Inconel 625

3D Printing Method: Selective Laser Melting (SLM)

The vast majority of metal 3D printing technologies utilize metal powder. As a result, the major differences between types of metal printers relate to how they fuse the powder into metal parts. Powder Bed Fusion is currently the most common type of metal 3D printing. These machines distribute a fine layer of powder over a build plate and selectively melt a cross section of the part into the powder layer. There are two distinct types of powder bed melting techniques: Selective Laser Melting and Electron Beam Melting. Out of these two based on the mechanical and surface finish requirements SLM method was selected to 3D manufacture the desired part.

SLM as known by many names such as Direct Metal Laser Sintering (DMLS), Selective Laser Sintering (SLS), Direct Metal Printing (DMP), or Laser Powder Bed Fusion (LPBF), most Powder Bed Fusion machines are actually Selective Laser Melting (SLM) machines. These use high-powered lasers to fuse

metal layers, after which operators remove and post-process the final parts. SLM is the current industry standard for metal additive manufacturing, with most companies selling SLM machines.

Raw material data sheet for the approved alternative material Nickel Alloy 625 is detailed below Fig.

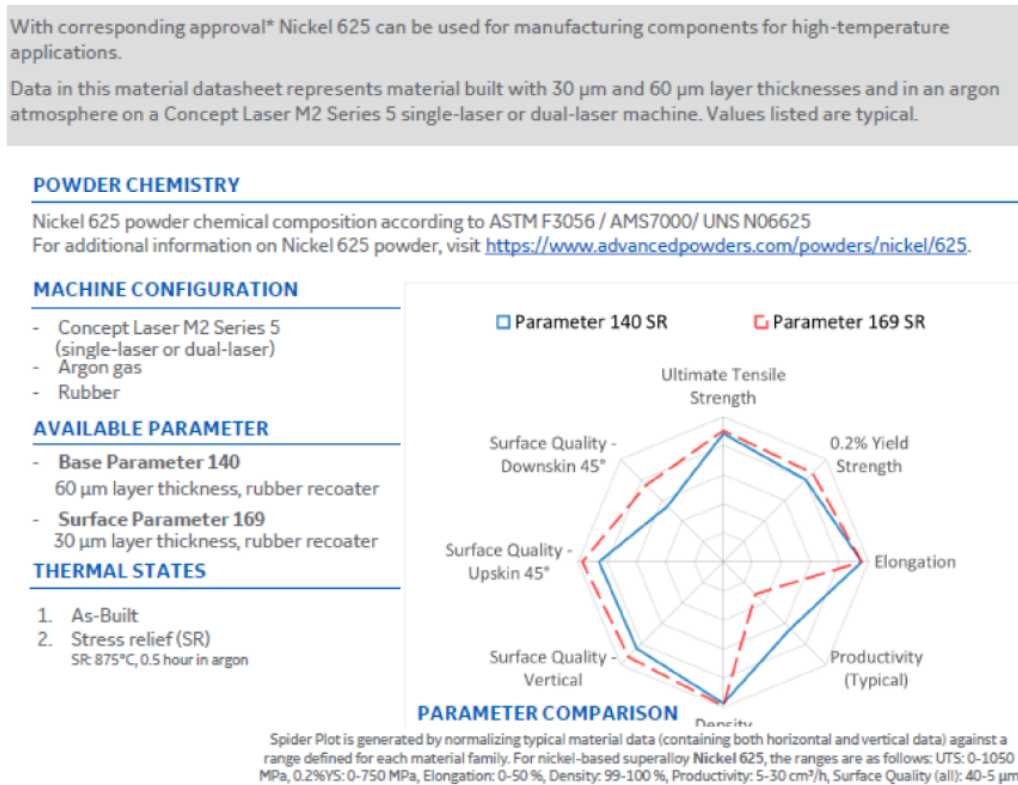


Figure 3—Raw material data sheet for Nickel Alloy 625

Digitization

The initial phase of the project involved digitizing and selecting six high-value spare parts, which served as a Proof of Concept (POC) for large-scale digitization within Field Operator Company. The process proceeded through the following steps:

- a) Data capturing and collection: High-end 3D capturing equipment was used to accurately record material properties, part geometry, and other relevant characteristics.
- b) CAD Modeling and 2D drawings: CAD modeling was carried out to generate detailed 2D drawings for all received parts, ensuring precision and compliance with required specifications.
- c) Development of Manufacturing and Testing Specifications (MTS): Comprehensive MTS documents were developed for each spare part, in accordance with international standards to guarantee quality and performance.
- d) Digital Library: Digital assets, including CAD models, 2D drawings, and MTS documents—were uploaded to DIS-RT, laying the foundation for field operator's digital warehouse of spare parts and enabling efficient management and accessibility of these digital assets.

Additive manufacturing

Additive manufacturing of the spare parts was completed within vendor's facilities, utilizing 3D printing for this project. The process followed these steps:

- Design for Additive for each part: Each part underwent a meticulous design process tailored for additive manufacturing, ensuring optimal performance and efficiency.
- Allocation of 3D printers: The selection of 3D printers is based on the size, geometry, and material requirements of each part, ensuring optimal printing conditions. SLM (Selective Laser Melting) machine/printer was selected for the intended part.
- Preparation of build files and orientation: Build files were prepared, and the orientation of each part was optimized for printing efficiency. Testing coupons were included in the building to ensure quality control.
- Heat treatment plans and cycles: Heat treatment plans were developed according to international standards to enhance the mechanical properties and durability of the printed parts.
- Post-production activities: Machining processes were employed to achieve the required properties and tolerances, ensuring the parts met precise specifications.
- Coating: Parts underwent coating processes as required, enhancing their corrosion resistance, surface finish, and overall performance.

Each step was meticulously executed to ensure the highest quality and functionality of the final printed parts.

Heat treatment

Stress relief/ Solution anneal at 875C to 1038C for 1 hour heat treatment cycle was applied in order to achieve the mechanical properties listed below:

Material	UTS (Mpa)	Yield Strength (Mpa)	Elongation at Break (%)	Hardness HV10
Inc 625 (ASTM F3056-14)	880	590	42	265

Quality Assurance and Testing

Adhering to the standards requires rigorous quality assurance and testing protocols to ensure the reliability and performance of 3D printed metallic parts. Key aspects of quality assurance include:

- Material Certification: Verifying that the materials used in 3D printing meet the specified standards, Data sheet requirement and are free from defects.
- Chemical analysis must follow a recognized national or international standard and use an appropriate method. Acceptance criteria should meet relevant design and material standards.
- Non-Destructive Testing (NDT): Employing techniques such as ultrasonic testing, radiographic testing, Liquid Penetrant testing and magnetic particle inspection to detect flaws and ensure the integrity of the printed parts.
- Mechanical Testing: Conducting tensile, impact, Hardness, Microstructure, Density Corrosion and fatigue testing to evaluate the mechanical properties of the printed components.
- Dimensional Inspection: Using precision measurement tools to ensure that the parts meet the required dimensional tolerances.

Rigorous testing and inspection are necessary to verify the integrity and performance of 3D printed part. ASTM F42 committee standards provide detailed methods for evaluating mechanical properties, dimensional accuracy, and surface finish.

Testing and inspections result for Visual, Destructive and non-Destructive test, heat Treatment, Impeller balancing as submitted by contractor are provided under Attachment-Quality control Report.

Regulatory Compliance

For the industries such as oil and Gas, regulatory compliance is of paramount importance. By following the guidelines set forth by organizations such as API, ASME, ISO, and NACE, additive manufacturing can be adopted to produce the components that meet the rigorous demands of the oil and gas sector.

Following the relevant standards, some of which are listed below, it ensures that printed components meet the requirements for production in the highly regulated Oil and Gas sector.

- API 20S - Additively Manufactured Metallic Components for Use in the Petroleum and Natural Gas Industries
- ISO/ASTM 52910: Additive Manufacturing—Design—Requirements, Guidelines, and Recommendations. This standard offers guidelines for designing parts for additive manufacturing, ensuring that they meet the necessary functional and performance requirements.

The impeller was tested in line with ASTM/API standards based on criticality ASM level-2, and the parameters of the testing are listed in the table below:

	Parameter	AMSL 1	AMSL 2	AMSL 3
Test specimens	Chemical analysis (see 7.4)	Feedstock certificate only	100 %	100 %
	Tensile testing (see 7.5)	1 specimen in 1 orientation (1 in Z) (1 total)	1 specimen in 2 orientations (1 in Z and 1 at 90° to Z) (2 total)	2 specimens in 2 orientations (2 in Z, 2 at 90° to Z) (4 total)
	Impact testing (see 7.6)	—	1 set in the Z orientation, 1 set 90° degrees from Z	1 set in the Z orientation, 1 set 90° degrees from Z
	Hardness testing (see 7.7)	—	100 %	100 %
	Microstructure analysis (see 7.8)	—	2 orientations (1 perpendicular to Z and 1 parallel to Z)	2 orientations (1 perpendicular to Z and 1 parallel to Z)
Component in the final condition	Visual inspection (see 7.10.1)	100 %	100 %	100 %
	Volumetric NDE (see 7.10.3)	Sample	100 %	100 %
	Surface NDE (see 7.10.2)	Sample	100 %	100 %
	Density testing (see 7.9)	—	—	Sample
	Hardness testing (see 7.7)	Sample	Sample	100 %
	Dimensional inspection (see 7.10.4)	Sample	Sample	100 %
Integrated substrates	Visual inspection for cleanliness on base material (pre-build) (see 7.10.1)	100 %	100 %	100 %
	Surface NDE on base material (pre-build) (see 7.10.2)	—	—	100 %

Figure 4—ASM Test parameters, Source: AM Vendor proposal PDO_MD MD03613

Benefits of Digital Assets/Digital Inventories

3D printing allows the parts to produce directly from a digital design file without requiring tooling, such as molds, which typically involve costs amortized over a minimum order quantity. In a digital inventory, numerous items can be stored along with their respective production and post-processing specifications.

When there is demand for a specific item, production can be initiated promptly, provided there is available capacity and necessary materials.

Additive manufacturing presents significant potential for enhancing supply chain management in the oil and gas industry. A primary advantage is the transition to digital inventory and on-demand production of spare parts, facilitating efficient inventory management. Through additive manufacturing, numerous spare parts, including those for legacy equipment, can be reverse engineered to establish a comprehensive digital inventory for on-demand fabrication, thus minimizing warehouse stock and associated storage costs. Over time, on-demand production and secure digital storage of component files may fundamentally reshape supply chains and distribution networks across the sector.

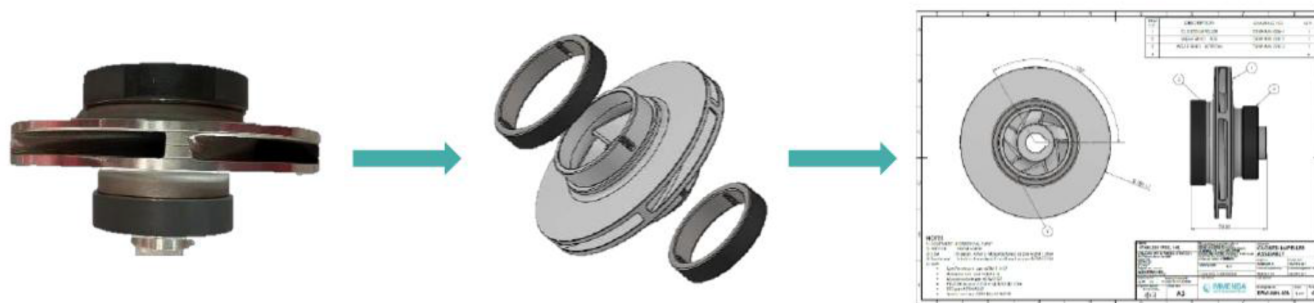


Figure 5—Study Part-Centrifugal Pump: Closed Impeller with 2 Wear Rings assembly (CAD File)

Conclusion

In conclusion, the implementation of Additive Manufacturing (AM) signifies a significant transformation within the Oil and Gas Industry, presenting substantial opportunities to improve operational efficiency, reduce downtime, and support sustainability objectives. The capability of AM to facilitate on-demand component production, streamline supply chains, and minimize material waste establishes it as a key driver of innovation in an industry characterized by high capital expenditures and logistical challenges. Additionally, AM's role in addressing environmental concerns and complying with evolving regulatory requirements contributes to enhanced corporate responsibility in today's dynamic market environment.

Nonetheless, effective integration of AM requires a comprehensive strategy that recognizes and addresses its inherent challenges. These include considerable initial capital investment, a shortage of skilled professionals, and complex regulatory landscapes, all of which necessitate targeted investments, workforce training, and inter-industry collaboration. Early adopters of AM are likely to achieve a competitive advantage as technology advances and its applications become more widespread throughout the sector.

Insights gleaned from trial component-Pump Impeller produced via Additive Manufacturing underscore its transformative potential while highlighting the imperative for systemic change to fully realize its benefits. AM is more than a technological enhancement; it serves as a catalyst for redefining operational models, enhancing resilience to market fluctuations, and advancing industry-wide sustainability efforts. As the Offshore Oil and Gas sector transitions towards cleaner energy and greater efficiency, AM is poised to play an increasingly critical role in shaping the industry's future, providing a clear avenue for sustained innovation and long-term competitiveness.

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